

Does Your EDR Do That?

Or: Not All EDR Products Have All The Features You'd Expect



- Chris Taylor
 - Founder and lead security consultant at Taksati Consulting
 - 25+ years of experience in digital forensics, incident response, threat intelligence, and so on
 - Builds and runs security programs for small media and entertainment companies
- Caveat emptor
 - You all know way more about Linux than I do
 - I will tell you the requirements and leave to you how best to meet them
- Any mention of a product by name is **NOT** an endorsement of that product, just illustrative examples

A Short Walk Through History...

TAKSATI CONSULTING
TECHNICAL SECURITY SERVICES

- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

- Anti-Virus (AV)
 - First Virus = Creeper infected a bunch of PDP-10s in 1971 (technically a worm, not a virus)
 - First Anti-Virus = Reaper by Ray Tomlinson, written specifically to remove Creeper
 - First use of the term “computer virus” = 1983 by Fred Cohen
 - First commercial AV product = Sophos in 1985
 - First commercial AV in USA = McAfee in 1987 (but that’s a different story)
 - First heuristic AV product = Flushot Plus or Anti4us in 1987, F-PROT in 1991 is first like modern
 - First virus on Linux = Staog in 1996
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

- Anti-Virus (AV)
- Anti-Malware (AM)
 - Virus = self replicating, appends to a file or system area
 - Worm = self replicating, travels via network usually using some exploit
 - Trojan = a stand-alone malicious file pretending to be safe, legit file
 - Rootkit = a collection of trojans and malicious kernel mods used to maintain access and hide
 - Every other type of malware is a Trojan defined by function rather than by type, e.g. 'backdoor'
 - Trojans are way more common than viruses, thus 'anti-virus' became 'anti-malware'
 - It now does a lot more than just AV (HIDS, DLP, etc), thus 'anti-malware' became 'endpoint security'
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
 - ~100,000,000 new malwares released every year
 - Signature databases got huge, always lagged, generally struggled to keep up
 - AV companies started leaning harder on the heuristics and behavior-based detections
 - First NGAV = Cylance released in 2012
 - There is no single standard definition of what “next gen” means
 - NGAV still relies on signatures, but heuristics engine is mo’ stronger
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)

A Short Walk Through History...

- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
 - First EDR = Carbon Black released in 2011
 - First use of term “EDR” = by Anton Chuvakin at Gartner in 2013
 - Originally designed to complement AV; it did NOT block malware, it enhanced response capabilities
 - Originally designed to compete with enterprise forensics, but be way faster
 - Collects ‘telemetry’ = logs every process start, network session, file access, etc.
 - Customers wanted less agents and less invoices... so NGAV and EDR merged
- Extended Detection and Response (XDR)

A Short Walk Through History...

TAKSATI CONSULTING
TECHNICAL SECURITY SERVICES

- Anti-Virus (AV)
- Anti-Malware (AM)
- Next Gen Anti-Virus (NGAV)
- Endpoint Detection and Response (EDR)
- Extended Detection and Response (XDR)
 - First use of “XDR” = by Nir Zuk of Palo Alto Networks in 2018
 - Full enterprise visibility requires monitoring logs, network, and endpoint
 - Those used to be three isolated silos
 - Combining them greatly enhances visibility and correlation of events
 - Tools that combine them call themselves XDR

EDR Market

THE FORRESTER WAVE™

Extended Detection And Response Platforms

Q2 2026



Which should I choose?

- It depends...
 - What are your needs?
 - What are your desires?
 - What is your budget?
 - What is your environment?
 - ...
- Choosing a product in four easy steps:
 - Step 1: develop measurable requirements
 - Step 2: compare options against those requirements, actually test them
 - Step 3: select the product that best meets those requirements
 - Step 4: figure out how to pay for it (that's a different presentation)

EDR Telemetry

TAKSATI CONSULTING
TECHNICAL SECURITY SERVICES

- <https://www.edr-telemetry.com/>
- A project by Defendpoint Consulting – I am not affiliated with this project in any way
- They tested a variety of EDR based on a standard grading criteria
 - Provides an apples-to-apples comparison
 - What telemetry do they collect?
 - What features do they have?

Telemetry Scores

TAKSATI CONSULTING
TECHNICAL SECURITY SERVICES

Rank	Windows		Mac		Linux	
	Product	Score	Product	Score	Product	Score
1	C-Prot	39.60	Phorion	35.00	C-Prot	25.60
2	Harfanglab	37.35	C-Prot	34.40	Uptycs	23.80
3	Uptycs	36.60	Elastic	19.80	CrowdStrike	16.95
4	Cortex XDR	35.80	CrowdStrike	14.60	Elastic	15.80
5	SentinelOne	35.75	MDE	13.70	SentinelOne	15.60
6	CrowdStrike	35.45	LimaCharlie	13.60	BitDefender	14.60
7	MDE	34.80	ESET Inspect	13.50	MDE	13.70
8	Elastic	33.45	BitDefender	12.90	LimaCharlie	12.70
9	ESET Inspect	30.80	Uptycs	12.30	Auditd	12.00
10	Trellix	29.60	Qualys	8.40	Cortex XDR	11.4

Notice how the scores differ by platform – the best Linux EDR scores lower than the 10th best Windows EDR

Telemetry Categories

TAKSATI CONSULTING
TECHNICAL SECURITY SERVICES

Windows

Process Execution

File Operations

Hash Algorithms

Network Connections

User Account Activity

Scheduled Task Activity

Service Activity

EDR SysOps

PowerShell Activity

Driver/Module Activity

Device Operations

Registry Changes

Named Pipe Activity

WMI Activity

BITS Jobs Activity

Miscellaneous System Changes

Mac

Process Activity

File Activity

File Metadata

Network Activity

User Account Activity

User & Session Activity

Scheduled Task & Persistence Activity

Service Activity

EDR SysOps

Script Activity

System Extension & Driver Activity

Process Tampering Activity

Access Activity

Device Activity

Code Signing & Trust Activity

Privacy & TCC Activity

Linux

Process Activity

File Manipulation

Hash Algorithms

Network Activity

User Account Activity

User Activity

Scheduled Task Activity

Service Activity

EDR SysOps

Script Activity

Driver/Module Activity

Process Tampering Activity

Access Activity

Telemetry Comparison

TAKSATI CONSULTING
TECHNICAL SECURITY SERVICES

		Linux			Windows		
		auditd	Carbon Black	CrowdStrike	sysmon	Carbon Black	CrowdStrike
Process Acty	Created	Y	Y	Y	Y	Y	Y
	Terminated	N	N	Y	Y	P	Y
	Tampered	Y	N	N	Y	P	Y
	Accessed				Y	Y	Y
	Loaded				Y	Y	Y
	Remote				Y	Y	Y
	Call Stack				Y	N	Y
	W32 API				N	Y	Y
File Access	Created	Y	Y	Y	Y	Y	Y
	Modified	Y	Y	Y	N	Y	Y
	Deleted	Y	Y	N	Y	Y	Y
	Opened				N	Y	Y
	Renamed				N	Y	Y
Network	Connection	Y	Y	Y	Y	P	Y
	Listening	Y	N	Y			
	DNS	N	N	Y	Y	Y	Y
	File Download				N	N	Y
	URL				N	N	N
Service	Created	N	N	P	N	P	Y
	Modified	N	N	P	N	N	P
	Deleted	N	N	N	N	N	N

auditd

- <https://github.com/linux-audit/audit-userspace>
- <https://man7.org/linux/man-pages/man8/auditd.8.html>
- Goal is to provide tamper-resistant audit logs that support security monitoring and incident investigation to make Linux compliant with Common Criteria, PCI-DSS, etc.
- Logs system calls, file access, authentication events, and configuration changes
- Uses configurable audit rules to monitor specific files, directories, users, processes, etc.
- Commercial EDR tools either
 - use it for a portion of their telemetry gathering or
 - require either tuning it or disabling it when installing their tool to prevent them from fighting

dnotify -> inotify -> fanotify

- Linux Kernel subsystem that provides monitoring of file system activity
- dnotify
 - monitors directory-level changes for file creation, modification, deleted, or renamed events
- inotify
 - monitors both files and directories and covers a broader range of events than dnotify
 - is optimized for tracking changes to specific files or directories, making it useful for file synchronization, configuration monitoring, and application automation.
- fanotify
 - provides system-wide file access monitoring and can intercept file open and access events.
 - is designed for security and antivirus use cases, allowing applications to inspect and optionally block file access before it occurs.
- Compared to inotify, fanotify provides broader visibility across filesystems and greater control over file access decisions
- Many commercial EDR and AV products use this for file access monitoring and blocking

Netfilter and nftables

- Linux Kernel subsystem that provides monitoring of network activity
- Netfilter is the engine within the kernel // nftables is the interface to configure that engine
- iptables is the legacy interface that has been replaced by the more modern and efficient nftables
- Provides a host-based firewall and IDS that can monitor and block network activity
 - Doesn't provide the IDS... it provides the IDS with efficient access to the traffic, something else provides detection rules
- Many commercial EDR products use this for network monitoring and blocking

- <https://ebpf.io>
- Extended Berkeley Packet Filter, a Linux Foundation Project, first extended in 2014
- Wraps programs to monitor system and kernel events to provide observability, networking, performance monitoring, and security
- Provides low-overhead, real-time visibility into processes, system calls, file activity, and network connections without requiring traditional kernel modules.
- Offers improved performance, flexibility, and compatibility compared to legacy monitoring mechanisms such as auditd, ptrace, and loadable kernel modules
- Several commercial EDR tools use this as a primary source for their telemetry

osquery

- <https://osquery.io>
- Developed by Facebook, first released in 2014
- Goal was to enhance endpoint monitoring and make enterprise-wide searches easier
- Exposes low-level OS system data as if it was a SQL database
 - Uses SQL language to return data from endpoints
 - `SELECT process.name, process.pid, cmdline FROM processes WHERE path LIKE 'C:\Users\%';`
- Facebook and Linux Foundation created a new foundation to host and control its code base
 - Open sourced all the code
 - Plugin framework allows it to be customized and extended
- Several commercial EDR tools are a custom frontend on osquery or use it for searching

- <https://github.com/microsoft/SysmonForLinux>
- Developed by Microsoft, ported to Linux, released as FOSS
- Generates structured event logs for activities such as process creation, network connections, file modifications, and module loading.
- Uses customizable XML-based configuration rules to selectively capture high-value security events.
- Integrates with SIEM and security monitoring platforms to support threat detection, incident response, and forensic investigations.
- Provides telemetry similar to Windows Sysmon, enabling consistent monitoring across Windows and Linux environments.

In summary...

When it comes to EDR features:

- Selecting a tool isn't just which is best
 - you want the one that best for your specific requirements
- The feature disparity between Windows and Linux is huge, and between tools is notable
 - Make sure the tool you select has the features you need
- Tools only have to partially support one feature to claim support for a whole category
 - Don't assume, test
- It's good to know what the “under the hood” components are your tools are depending on
 - Supply chain risk management, software bill of materials

Fin

Chris Taylor
Taksati Consulting
info@taksati.org